Construct AI: An Integrated AI-Driven Framework for Automated Pre-Construction Feasibility Analysis and Decision Support

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***Abstract*— Evaluating the feasibility of a construction project continues to be contingent upon a series of independent specialist studies. The surveyor grades the plot, the architect makes up plans, a consultant acquires necessary statutory approvals, and the quantity surveyor prices out the project, using different methodologies and never comparing the outcome to other results. This paper introduces ConstructAI, a tool which combines all of the above studies into one automated pre-construction study. Four integrated modules update a unified project data model: a geospatial module, which uses optical character recognition and YOLOv8 for interpreting survey plans, and matches them to satellite terrain and flood information for grading the plot; a validation module, which recognizes walls, openings and columns for deriving area, volume and quantity of elements; a compliance module, which performs validation of geometry against by-laws of the Urban Development Authority and provides a sequence of approval pathway with a penalty risk attached to it; and a cost and lifecycle module, which uses a model trained on past workbooks for generating a bill of quantities, then pricing it using market rate scraping and predicting service life and maintenance cost through an XGBoost regression with thirteen factors. In place of an interactive assistant, the system offers a visual audit report and a scenario simulator with sliders. We present the architecture, the data exchange between the modules, and the evaluation protocol adopted for implementation in Sri Lanka. Component evaluation is in progress; we report the architecture and the evaluation protocol, not measured performance.**

Keywords— ***pre-construction feasibility, construction decision support, YOLOv8, automated compliance checking, XGBoost, bill of quantities, building service life, geospatial analysis***

I. Introduction

n most cases, virtually all aspects of the building's cost are determined prior to breaking ground. The money that is to be spent on plot, walls, grade of concrete for the walls to be cast and the maintenance regime that the client will pay for are all committed money which cannot be recovered easily at a later date. However, studies used to support the decisions are still conducted separately. Each step creates documents instead of data, and each handover is a chance to re-enter the information manually. Since many projects have been carried out in Sri Lanka, it has been invariably observed that there have been some mismatch between the cost and time estimated during the initiation phase of a project and the actual cost and time spent during implementation [1].

This is important since the separate studies are not independent. For example, a piece of land that gets flooded twice within ten years results in change of foundation type, and therefore, change in bill of quantities and thus risk for the client in case the permit is denied. Violation of setback rule when obtaining permits means that the previously priced bill of quantities becomes null and void. These conflicts get sorted out through consultations and drawing modifications and this is costly to the extent that it is rarely done.

GIS with machine learning approaches assign weights to site factors and produce suitability maps [2], [3]. Detection algorithms recognize elements of architecture in plans [4], [5]. Semantic and knowledge graph approaches encode the building code in order to enable automated checks of compliance with it [6]–[8]. Regression and ensemble modeling predicts costs of construction based on its early design characteristics [9]–[11]; data-driven modeling predicts durability of reinforced concrete buildings [12]. What is still lacking is the combination of all these processes. The suitability map created with one system cannot be processed by another system, and measurements made with takeoff software have to be manually inputted into spreadsheets for estimation of prices. Similar conclusions were reached in review papers about decision support systems in construction engineering [13].

ConstructAI is the name of the approach proposed in this paper, in which preconstruction feasibility evaluation is considered as one analysis instead of four. The contributions of our work can be summarized as follows. First, we introduce a four-component architecture in which all components are working on a single project data model so that changes in site assessment can automatically affect cost and life cycle forecasting.

Secondly, a pipeline for recovering a georeferenced boundary from a scanned survey plan and comparing it to the satellite terrain and flood data is described, which grades the boundary. Thirdly, we merge the bill of quantities generator with real market prices and a thirteen-parameter service-life regressor: this way, capital cost and fifty-year maintenance cost are generated from the same set of inputs. Fourthly, the conversational assistant used in such systems is replaced with

a visual audit report and a scenario simulator, an interaction paradigm that is suitable for users who are interested in seeing the consequences rather than reading them. Finally, the framework is instantiated for the Sri Lankan regulatory setting, where Urban Development Authority by-laws and local material prices govern what is buildable and what it costs.

II. Related Work

A. Site suitability from geospatial data

The multi-criteria evaluation in the context of a geographic information system is the standard method for evaluating lands for a certain application, and new research has used weights based on data rather than expert-provided weights for criteria, which reduces subjectivity inherent to the classic methodology [2]. Cloud computing technology allows for routine access to the underlying imagery: Google Earth Engine provides a platform for accessing decades-long archives of satellite imagery and terrain layers through a framework of analysis, and thus such factors as elevation, slope, land cover, and inundation history become accessible for any polygon regardless of local data administration [3]. Methods for the detection of vacant lands and their construction suitability have been based on combining the mentioned imagery with parcel information for the identification of developable lands within cities [14]. These methodologies answer the question of whether a particular location is suitable in general terms; however, they do not answer whether a particular building can be placed there.

B. Understanding architectural drawings

Structural element detection from floor plans is a mature computer vision problem with a difficult long tail. There is a huge variation of scales, ranging from the size of a door panel to a building silhouette, and there are differences in drawing conventions across offices. Multiscale detection systems have been specifically designed for complicated architectural drawings with such a scale variance [4], and pipelines for full-floor plan parsing that include both detection and vectorization have been described [5]. YOLO framework, which solves object detection through a grid-based regression [15] and its eight revision implements anchor-free object detection with an efficient training pipeline [16], is a standard backbone for this problem since the inference speed is sufficiently fast to be performed interactively on uploaded drawings. However, in most papers the research is limited by structural element detection. Converting those recognized elements to quantities consumed by the cost model involves scale recovering and unit calculation, which is rarely covered.

C. Automated regulatory compliance

Automated code checking has been worked on for two decades now, and the hard part of the problem has always been not checking but encoding, i.e., turning building regulations from prose into something executable. Semantic web ontologies have been employed both for building the representation of the building model and the regulation in the same vocabulary so that inference can be performed about both [6], and knowledge graphs have been utilized to bind regulatory provisions to entities in the model more robustly than in handwritten rule scripts [7]. Review articles in the field point out that the adoption of automated code compliance checking has been constrained by the rapid turnover in the regulations as well as the maintenance burden of the encoded rules base [8]. In the case of national deployment, the actual challenge is more limited: a small number of important numerical rules, like setback distances, floor area ratio and road widths, account for most of the refusals of a small project.

D. Cost, schedule and lifecycle prediction

A lot of research effort has gone into parametric and learned cost models, and in both cases tree ensembles significantly outperform classical regression at the early stages of estimating the cost of buildings, when the number of features and relations between them is rather small [9]. There have been hybrid and stacked models of construction cost estimation using random forests and gradient boosting [10], and there have been applications of interval estimation in estimation of labor unit costs, which vary much more than material costs [11]. The automated quantity surveying problem has been also approached from the perspective of web applications to decrease the arithmetic mistakes made in manual preparation of bills [17]. The durability problem has been attacked by machine learning approaches to estimating service life and deterioration of reinforced concrete structures, when the environmental exposure determines the results [12]. As for scheduling, there have been learning-based solutions to assembly planning in prefabricated construction [18]. However, these lines of research have rarely been joined, meaning that a customer is informed about the cost of building construction but not maintenance.

E. Positioning

Table I compares ConstructAI with representative prior work. The difference here is not about the uniqueness of any particular component, but about the fact that all four analyses are performed on the same representation, which ensures consistency between their results.

Table I. COVERAGE OF REPRESENTATIVE PRIOR WORK

Approach	Site	Plan	Permit	Cost / lifecycle
GIS multi-criteria suitability [2], [14]	Yes	No	No	No
Floor plan parsing and detection [4], [5]	No	Yes	No	No
Automated code compliance [6]-[8]	No	Partial	Yes	No
Learned cost estimation [9]-[11], [17]	No	Partial	No	Cost only
ML durability assessment [12]	Partial	No	No	Lifecycle only
ConstructAI (this work)	Yes	Yes	Yes	Both

III. System Architecture

The ConstructAI system is structured in five layers which are depicted in Fig. 1. The first layer takes as input the survey plan, architectural plan, the site location and some design parameters. In the data acquisition layer, the input is processed: the document images are processed by the optical character recognition technique, the site location is used for querying the services of terrain and imagery, and regulation knowledge base and market rate data source provide the reference data on which the analysis is compared. In the intelligence layer there are the four components which are denoted by C1 to C4 in figures and tables and referred to by their full name in Table II: C1 analyzes the site, C2 validates the plan, C3 provides the approvals and C4 deals with the cost and lifecycle of the project.

The integration layer is the layer that makes the framework something other than four tools under one login. Each of the components operates on a single document made up of four sections: a site section containing the parcel polygon and the environmental characteristics calculated for the parcel, a geometry section containing the geometrical elements reconstructed from the drawing, including the scale at which they were reconstructed, a compliance section containing the rules examined and their results, and a cost section containing the costed elements and life cycle analysis. Since the geometry section is only written once, but read by both the compliance and the cost components, an alteration of a wall thickness must not result in the compliance decision being different from the bill of quantities in terms of the same building. Table II describes the roles of the components.

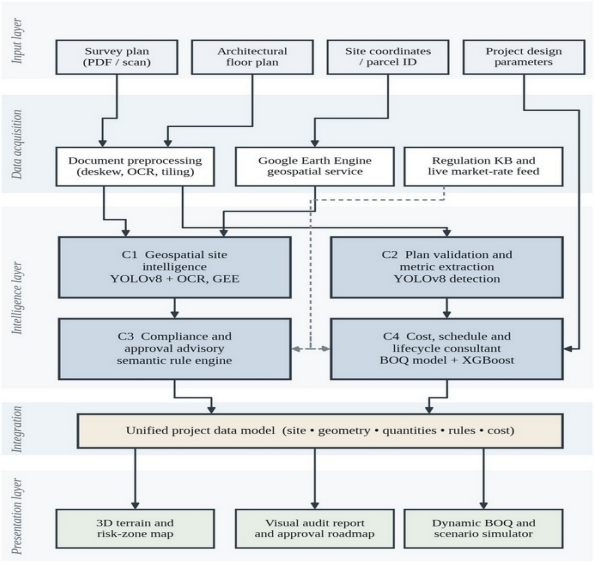


Fig. 1. Layered architecture of the ConstructAI framework. Dashed edges carry reference data rather than project data.

Table II. COMPONENT RESPONSIBILITIES, TECHNIQUES AND PRINCIPAL OUTPUTS

Component	Core function	Principal techniques	Written to shared record
C1: Automated Site Analysis and Geospatial Intelligence Module	Grade the parcel for geographic and environmental suitability	OCR, YOLOv8 boundary detection, Earth Engine queries	Parcel polygon, buildable and risk areas, terrain
C2: Automated Architectural Plan Validation and Metric Extraction Module	Recover quantities from the architectural drawing	YOLOv8 detection, annotationbased scale recovery	Element counts, areas, volumes, by-law flags
C3: Automated Regulatory Approval and Compliance Advisory Module	Establish the approval pathway and the exposure of deviating	Parameterised rule predicates over the shared record	Approval roadmap, progress checklist, risk penalty matrix
C4: Intelligent Cost, Scheduling and Lifecycle Consultant Module	Price the works and forecast long-run performance	Learned BOQ generation, rate scraping, XGBoost regression	Costed bill of quantities, schedule, service life profile

IV. METHODOLOGY

A. *Component 1 (C1): geospatial site intelligence*

The pipeline of the site, illustrated on the left side of Fig. 2, starts with the uploaded survey plan, which in reality is usually an image or a scan of the document created years back. The optical character recognition technique retrieves the bearings, distances, and lot information annotated on the plan, and the YOLOv8 detector finds the boundary polyline and the annotation boxes, such that text is linked to the corresponding boundary. The retrieved traverse is then closed and georeferenced using the given coordinates to generate the parcel polygon.

This polygon serves as the ROI for a series of queries in Google Earth Engine from which we receive information on elevation, slope, historic inundation and land cover. All of this information is synthesized into what we call a "buildable area" mask - anything over a particular slope value, falling within an historic flood extent or inside a reservation is disregarded, and the rest is outputted as "usable area" along with a three-dimensional map of the terrain. This same query also provides three of the lifecycle inputs, distance to the coast, elevation and industrial land, which is one of the data dependencies.

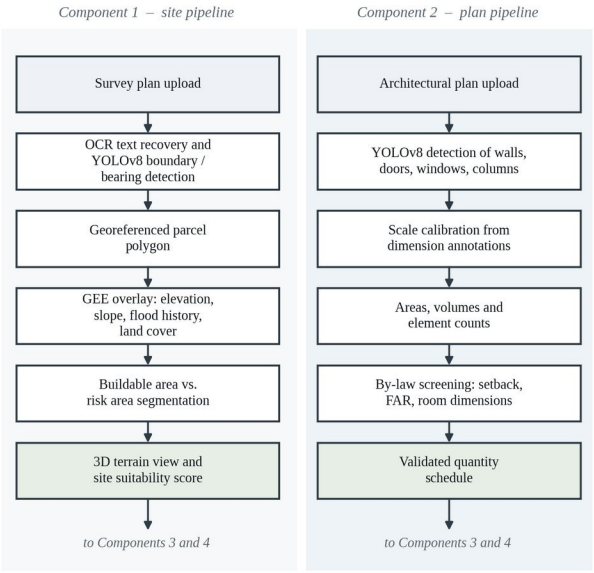


Fig. 2. Document-understanding pipelines of Components 1 and 2. Both terminate in structured records written to the shared project data model.

B. *Component 2 (C2): plan validation and metric extraction*

On the other hand, the plan pipeline, illustrated on the right-hand side of Fig. 2, locates walls, doors, windows, and columns in the uploaded architectural plan. Detection by itself does not suffice for the derivation of quantities, since a bounding box can be represented in terms of pixels. Thus, a plan pipeline extracts a scale factor from the dimensions annotated in the plan in terms of the recognised text and the pixel distance between the extension lines used in the annotation of the dimensions, defaulting to an explicitly defined plan scale in case there are no such annotations or in case they cannot be correctly recognised.

The recovered geometry undergoes testing against the dimensional bylaws once it is out of the component in order to highlight any apparent issues, such as having a room smaller than the minimum dimension or a building footprint

encroaching on the necessary setback, while the designer is still viewing the drawings instead of after costs have been determined.

C. Component 3 (C3): compliance and approval advisory

Compliance consists of comparing the shared project record with a subset of the UDA regulations and local authority by-laws encoded as parameterised predicates over the site and geometry sections. Thus, changing the permitted floor area ratio is equivalent to changing a value and not the checker itself. The implemented rules and the areas outside coverage are listed in Table V. There are three artefacts generated. The approval roadmap is a list of the required permits in sequence of obtaining them along with the consumed documents and indicative processing times. The progress tracker is a derived checklist from the approval roadmap updated by the user as approvals are obtained. Finally, the risk penalty matrix indicates, for each rule that is unsatisfied, what is the statutory cost of proceeding in defiance of it, i.e. the information the client needs to make the decision.

The component itself is rendered as an "auditor in sight" as opposed to the interface that would enable chatting. The component generates the report that contains an annotation of each finding that corresponds to the part of the diagram or location where it occurred and the clause used to generate the finding. The voice-based query interface that enables asking of the regulatory question is laid on top for those who want to ask the question, but the answer will be based on the same rule base that generated the report.

D. Component 4 (C4): cost, schedule and lifecycle consultant

Component 4, as shown in Fig. 3, converts the geometry and design inputs into financials. There are two paths that go in parallel. The cost path takes the quantities written by Component 2, or a drawing uploaded directly to Component 4 when a user wants an estimate without the full analysis. These quantities are fed to the bill of quantities generator, which is trained on the workbooks of previous projects. This bill of quantities generator does not use a template but knows what to do with the geometry. For example, a raft foundation would automatically include all the excavations and blinding and reinforcing that goes along with it. The rates used in the bills are scraped from the published prices of cement, steel, sand, and labor.

The same priced bill also drives an indicative schedule. Each item is mapped to an activity type and its quantity converted to a duration through a productivity rate; the activities are ordered by the conventional sequence of substructure, superstructure, roofing, finishes and services, and the priced amounts are distributed over that order to give a phased cash-flow profile, so the client sees when the money is required and not only what the works cost. The schedule is indicative rather than resource-levelled, and is not evaluated separately in this paper.

The lifecycle path uses thirteen variables identified in Table III to forecast the service life and a maintenance cost path through a gradient-boosted tree ensemble [19]. Tree ensembles are appropriate in this case since the relationship is extremely non-linear and interactive: proximity to the sea is highly significant for low concrete grade but not very significant for a high concrete grade, and the impact of humidity varies based on wall thickness. The maintenance cost path is represented in terms of an annual cost path with a depreciation formula applied to the resulting cost value in order to make the forecast summable over a fifty-year period and comparable to the capital cost derived from the other path.

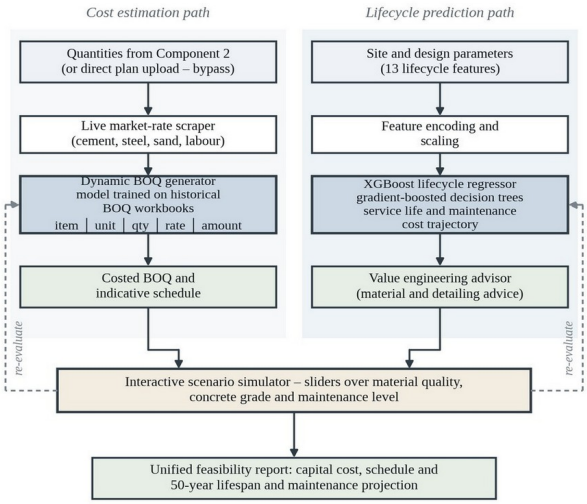


Fig. 3. Component 4. The cost and lifecycle paths consume different inputs and are reconciled in the scenario simulator.

Table III. PARAMETERS CONSUMED BY THE LIFECYCLE REGRESSOR

Group	Parameter	Type or range	Role in the prediction
Environmental	Distance to sea	metres	Chloride-driven reinforcement corrosion
	Soil pH	pH units	Chemical attack on the foundation
	Average humidity	per cent	Moisture ingress and wall deterioration
	Annual rainfall	millimetres	Load on external wall finishes
	Industrial proximity	kilometres	Airborne chemical exposure
	Land elevation	metres	Drainage and retained ground moisture
Structural	Concrete grade	G20, G25, G30	Compressive strength and cover
	Material quality	1-5 scale	Quality of sand, aggregate, steel
	Workmanship quality	1-5 scale	Execution quality on site
	Wall thickness	4.5 in or 9 in	Envelope thermal and moisture buffering
	Foundation type	pad, raft, strip	Settlement under imposed load
	Number of floors	count	Load transferred to the substructure
Maintenance	Maintenance level	1-5 scale	Intervention frequency, with depreciation rule

The two advisory functions are placed above the forecasted results. The value engineering advisor reviews the parameter vector for combinations that are known to reduce

the lifespan of the structure and makes a recommendation for a substitution. For instance, the advisor may recommend using epoxy coated reinforcement in case the site is located near the coast and shows how much maintenance cost the substitute would save. The scenario simulator represents the adjustable parameters as sliders where the user may decrease the quality of materials and the maintenance and see the cost of the construction decrease and the cost of the 50 year period increase.

Interaction design

It is worth stating the interaction choice explicitly, because it departs from common practice. Typically, systems of this type would have an interface featuring a chatbot. The use of a chatbot in conducting the feasibility analysis is impractical, as the user would not know how to formulate questions, the answer cannot be linked to any drawing, and the origin of the produced answer cannot be identified. ConstructAI employs two interfaces that do not have conversational properties. Every finding in the visual audit report can be linked to the corresponding location and the particular clause. Scenario simulator allows users to modify the decision made to see its impact on the result.

Failure modes of the two surfaces are different, too, and that forms another reason why the two surfaces are good. An audit report states what input is missing when the rule cannot be assessed. A simulation gives an extrapolation when the system is asked about a combination of inputs that falls beyond the training scope. A generated reply has no comparable failure mode: it reads as fluent whether or not the analysis behind it succeeded.

V. IMPLEMENTATION AND DATA

These components are trained offline and hosted behind a single web client, where the project’s shared history is stored per project such that the user can revisit the analysis once the approval has been obtained or the drawing reworked. The survey plan detector and the architectural plan detector are both trained with YOLOv8 models on annotated drawing datasets, and the geospatial queries are issued against Google Earth Engine; the lifecycle regressor is based on an XGBoost model. The dataset used to train the bill of quantities generator is a large bill workbooks dataset gathered from practising quantity surveyors, while the lifecycle model is trained on a large dataset of buildings with thirteen parameters and observed condition annotations. Dataset curation is ongoing and the corpora continue to grow, so the evaluation procedure is presented below in place of final dataset statistics. Table VI lists the configuration items that must accompany those statistics.

The evaluation of all the models is done using a project-disjoint split, such that none of the drawings and workbook from projects included in the test set appear in the training data. This is important as it turns out; because of how bills of quantity from one practice have the same wording and format, using a random split on items will lead to the model memorizing the format.

Shared record is versioned and not overwritten. Then each execution generates a new revision with the model versions and a snapshot of market rates at the time of the execution so that an estimate can be regenerated months later after new market rates have been set and new detectors have been trained. This is important in relation to a report that will be used by the client for seeking funding, where it is not only the estimate that is important but the basis of the estimate. It also provides the evaluation with a history of the review, since each reported metric has a revision and model checkpoint identifier associated with it.

VI. EVALUATION PROTOCOL AND ACCEPTANCE TARGETS

Each of the parts is evaluated based on its individual score relative to its objective, rather than an aggregate measure. The detection accuracy for Parts 1 and 2 is measured in terms of mean average precision at the intersection-over-union threshold of 0.5 and with the error in the computed quantities, because the detector could perform quite well but provide a poor area measurement if the scale cannot be recovered. Part 3 is evaluated in relation to the drawings provided by the chartered professional in terms of rule-level precision and recall, to ensure that missed violations and false positives are counted separately. Part 4 is reported as mean absolute percentage error compared to the costed bill created by the quantity surveyor for the project.

Table IV below outlines the measures chosen along with the threshold level established for each measure. These threshold levels have been set before training on the basis of what the practitioner would accept in lieu of the manual process that is to be automated; for instance, a number that differs by only a few percent from that of the surveyor's take-off number is acceptable but not a compliance checker which fails to identify one out of ten violations. These levels have thus been established as goals which will be assessed at the end of the evaluation.

Table IV. EVALUATION METRICS AND ACCEPTANCE TARGETS

Component	Task evaluated	Metric	Target	Status
C1	Boundary and annotation detection	mAP@0.5 / boundary IoU	0.90 / 0.85	In progress
C1	Buildable area delineation	Area error (%)	≤ 5	In progress
C2	Element detection	mAP@0.5	0.90	In progress
C2	Quantity extraction	Area and volume error (%)	≤ 4	In progress
C3	Rule evaluation against adjudicated plans	Precision / recall	0.95 / 0.90	In progress
C4	Bill of quantities generation	MAPE against priced bill (%)	≤ 10	In progress
C4	Service life prediction	R^2 / MAE (years)	0.85 / 3.0	In progress

VII. DISCUSSION AND LIMITATIONS

Several limitations arise from the implementation. The pipeline of site is as good as the survey document it receives, and old Sri Lankan survey documents tend to be faded, handwritten, and photographed at a tilt; failure in recognition of such documents affects every downstream operation, which is why the parcel polygon is shown prior to analysis to be verified. The compliance checker implements only a limited set of the applicable laws, which laws are subject to change; the database of rules is kept up to date in data form because of that, but it needs a curator nonetheless. The price scraper relies

upon the ongoing availability and format of the documents it extracts prices from; it is more harmful if the rate does not update than when it is just not present, so the interface shows the age of every rate.

The lifecycle approach comes with the biggest caution: fifty-year forecast cannot be tested by fifty-year out-of-sample data. As it is trained on the observations of condition and maintenance cost for the periods for which there are records, its forecast for long periods needs to be seen as an extrapolation based on the given depreciation rate assumption rather than as a measurement. The approach itself is also localized for one jurisdiction; geospatial and drawing tools can be transferred without much adaptation, while the compliance rules and rates sources are jurisdiction-dependent.

Set against these limitations is the kind of user the framework is built for. A large developer already retains a surveyor, an architect and a quantity surveyor, so automating their coordination adds little. A first-time homeowner or a small contractor retains none of them and proceeds on little more than guesswork; for that user the comparison is not with a professional team but with no analysis at all, and an estimate within ten per cent, delivered before any money is committed, is worth more than a precise estimate delivered after the foundation has been poured.

VIII. CONCLUSION

In this paper, an integrated framework for pre-construction feasibility analysis was set out, in which site evaluation, drawing verification, compliance checking, cost estimation and lifecycle prediction are performed against a single shared representation of the project. The contribution lies less in any individual model than in their composition: because all four components read and write the same record, a revision in one analysis is reflected in the others without manual re-entry, and the client can be given a capital cost and a long-run maintenance cost derived from the same geometry. The interaction design deliberately excludes a conversational front end, because a feasibility decision needs provenance and sensitivity rather than fluency. The components are implemented and the protocol of Table IV is defined, but measured results are not yet available; what is reported here is the architecture and its evaluation design, not validated performance.

Research work in progress focuses on concluding the assessment described in Table IV, expanding the set of rules from the by-laws to include dimensions other than those dominating the rejections for smaller projects, as well as incorporating the condition information gathered from the instrumented buildings in the life cycle model so as to make its projections measurable.

APPENDIX

Table V. IMPLEMENTED BY-LAW RULES

Rule	Parameter in the prototype	Evaluated against
Front setback / street line	Setback distance (m) for the zone	Footprint vs. parcel polygon
Side and rear space	Clearance (m) to each boundary	Footprint vs. parcel polygon
Floor area ratio	Permitted FAR for the zone	Total floor area / plot area
Plot coverage	Maximum coverage (%)	Footprint area / plot area
Building height	Maximum storeys for the zone	Storey count from the plan
Access road width	Minimum width (m) at the frontage	Frontage read from survey plan
Minimum room size	Minimum dimension (m) and area (m²)	Room polygons recovered by C2

Source: UDA Planning and Development Regulations; room size from local authority by-laws. Outside coverage: structural, fire, drainage, environmental and heritage rules.

Table VI. CONFIGURATION REQUIRED FOR REPRODUCIBILITY

Item	Value	Item	Value
Survey-plan detector variant	[fill]	BOQ corpus: workbooks and period	[fill]
Survey-plan training images / source	[fill]	BOQ item taxonomy and matching rule	[fill]
Plan detector variant	[fill]	Rate sources and refresh interval	[fill]
Plan training images / source	[fill]	Lifecycle corpus: records and horizon	[fill]
Annotation protocol and classes	[fill]	XGBoost hyperparameters	[fill]
Train / val / test split (projects, images)	[fill]	Feature preprocessing, missing data	[fill]
Input size, epochs, batch, optimiser, LR	[fill]	Hardware and framework versions	[fill]
OCR engine, preprocessing, scale fallback	[fill]	Random seed and number of runs	[fill]

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